AdaptEngine — Week 7 Report (Group 15)

Version: 0.3.0 · **Date:** 9 November 2026 · **Evidence file:** `docs/WEEK7_REPORT_NOTES.md` (every number below appears there with its command or results file) · **Full tables:** `docs/WEEK7_RESULTS.md`

Week 7 theme: *make the evaluation honest, close the loop, stress-test, then demo.* Assignment-2 W7 items covered: (1) mismatched simulator variants and stress cases, (2) sensitivity over heuristic parameters, (3) measured improvements with limits. The optional ranking policy is deferred to W8.

---

1. Summary of changes and why

Change	W6 defect it fixes	Notes ref.
Executed-state prerequisite audit (primary safety metric, D46)	W6 audit assumed every answer was correct, hiding 183 (adapt) / 640 (static) violations in calm	T1.1
Closed loop: `Engine.next_activities` (replan each day and after every wrong answer), mid-day budget, exercise repeats (D41), best-effort plans (D42)	The harness never replanned after outcomes; learners sat idle (50 idle days in calm)	T2.1–T2.4
Reachable spaced review: configurable half-life, maintenance tier, single recall implementation (D43)	Reviews could never fire; adapt = no_review everywhere	T2.6
Diagnostic scored with Brier/log-loss/AUC vs binary truth; flags (min-k, soft pruning, frontier, population prior); pre-registered selection rule applied as written (D44)	MAE of p against θ was an improper metric; W6 overrode its own selection rule	T3.1–T3.3
Simulator worlds S0–S4 and a pre-registered stress grid (D47)	One simulator that rewards raw minutes; results interpreted after the fact	T4.1–T4.4
Exactly-one trace audit; oracle records superseding greedy ones	45 SELECTED records for 21 activities on the oracle path	T1.4
Real pyBKT parity, BKT fits on ASSISTments 2009 and Cognitive Tutor (D48)	Parity test never used the library; parameters never fitted	T1.5, WP7
Precomputed graph indices	`descendant_count` recomputed by NetworkX in the scheduler's inner loop	T6.1
Demo frontend (FastAPI adapter + React; engine is the sole decision source, D45)	—	WP8

2. Corrections to W6

W6 claim	Corrected
0 prerequisite violations for adapt	183 (calm), 152 (missed_day) executed violations on W6 code; **0 in W7**
static 0 violations in calm	640 executed violations (87/100 learners)
Adaptive diagnostic quality (MAE)	Brier vs truth 0.2425, worse than using no diagnostic (0.2080)
"Assessed-only parity"	0.3852 vs fixed@n 0.2555 — not parity
Diagnostic defaults h_stop 0.35, cap 2	Code was h_stop 0.50, cap 3, N 8 (W7 rule: population prior 0.24, cap 2, N 6)
Items ≈ 2.7	2.96
Planner BKT "T 0.10, G 0.15, S 0.15"	p0 0.20, g(d) = 0.30(1 − d), s(d) = 0.05 + 0.25d, T_ex 0.05, T_lesson 0.20
Nominal simulator half-life 30 days	Nominal simulator has no forgetting (30 was the planner's value)
100% trace coverage on the oracle path	Duplicated/stale records (fixed, T1.4)
Results artifact = 100 seeds	Tests had overwritten it with a 5-seed run (fixed, T1.3)
Decision IDs D36/D38/D40	Mismatched with `DECISIONS.md`; D19 never existed (now recorded as reserved)
TC6.x in `tests/test_replanner.py`	Tests live in `tests/test_w5.py`

3. Closed-loop results (S0–S4 matrix)

Pre-registered (`docs/preregistration/W7_T4.1_2026-11-08.md`): 5 worlds × 3 curricula × 4 budgets × 8 scenarios (4 stress) × 5 arms, N = 200 paired learners per cell, headline N = 1000; 420 cells.

H1 (safety). The registered run *failed* H1: arms with spaced review still had executed violations (adapt 6192 summed over cells; no_review 0). Cause: due reviews and remediation bypassed the prerequisite gate (Defect 7). After two fixes, each with a regression test that fails on the previous code, the final grid has **0 executed violations for every adapt arm in all 420 cells**, against 1,649,720 for static and 1,229,249 for no_replan. Adapt also has 0 idle days (static 103,074, no_replan 150,429). All three runs are kept (`w7_grid_prereg.json`, `w7_grid_fix1.json`, `w7_grid.json`).

H2 (headline, S1 prerequisite-gated world, 4-node, B=15, calm, N=1000). Goal true-mastery 29.0% (adapt) vs 28.7% (static): Δ +0.003 [−0.008, +0.014] — parity, with 0 vs 7200 executed violations and all-required mastery 3.9% vs 1.8%. In S0 the same cell is −0.013 [−0.036, +0.010].

Across all cells: adapt is better than static on goal mastery in 35 cells, not significantly different in 226, worse in 159; on mastery of *all* required concepts it is better in 45 and worse in 21.

Where adapt loses (T4.5). (a) S0/S2 reward raw minutes on the goal whatever the learner's readiness; static and no_replan buy goal minutes with millions of violations. (b) After disruptions adapt rebuilds prerequisites first, so a 3–5-day horizon leaves fewer goal minutes. (c) In S4 (BKT world) adapt correctly stops practising concepts it has mastered, so goal-only mastery is lower in calm (−0.150) while all-required mastery is higher (+0.125); with a 21-day horizon it wins by +0.405. (d) Goal θ at equal minutes is negative in most cells: safety costs goal minutes.

H3 (spaced review, fast forgetter, 21 days, h = 7). Review changes the outcome in 32 of 60 cells: positive in 11 (e.g. 8-node +0.041), negative in 21 (synth-20, where reviews crowd out new content).

Expected-workload projection is worse than the optimistic projection in 113 cells and better in 9.

4. Diagnostic matrix and calibration

Held-out Brier vs 1[θ ≥ 0.70] (lower is better): prior-only 0.2080, W6 diagnostic 0.2425, pre-registered winner 0.2324 (Δ vs W6 −0.0101 [−0.0210, −0.0015]). Across worlds (250 learners): no BKT diagnostic beats the prior in S0 or S2 (TC2.1's equal-budget clause is **not met** there; fixed-order at the same item count is better), all beat it in S4, where the R§7.4 recommendation (min 2 items, no pruning, frontier) reaches 0.0589 (AUC 0.945). A per-concept IRT CAT is best in S0; a global-ability CAT is best in S2 by AUC. The state-space diagnostic and fitted-sim parameters under-perform. Which diagnostic looks best depends on which world generated the data.

5. Sensitivity

Scans of θ, k, half-life, p0, T_lesson, recall and remediation thresholds, a θ×k factorial and alternative mastery criteria (S0/S1, 4-/8-node, N=200): the engine is insensitive to θ in 0.70–0.90, to the thresholds and to h ≥ 14; strict evidence (θ = 0.95: −6.5 points; 3-in-a-row: −12.5 points in S0 4-node) costs goal mastery in a fixed short horizon. Executed violations are 0 in every configuration. Tornado charts: `docs/figures/w7_tornado_*.png`.

6. Measured improvements

Improvement	Before → after
Planning latency, identical plans (564/564)	synth-50 60.4 → 22.5 ms (2.69×); oracle cell 24.1 → 9.5 ms (2.55×)
Safety (4-node calm, executed violations)	183 → 0 (static 640)
Calibration, held-out RMSE (fitted vs heuristic)	ASSISTments 0.4746 → 0.4414; CT 0.4952 → 0.4549; simulated S0 0.5716 → 0.4586
Diagnostic Brier (S0 held-out)	0.2425 → 0.2324

Limits: fitted parameters are not adopted as defaults (maths data; they hurt the diagnostic), and the diagnostic still does not beat the prior in S0/S2.

7. Real-data fitting and provenance

ASSISTments 2009 (297,016 answers, 81 skills) and Cognitive Tutor (15,935 rows, 9 KCs): fitted priors are high (0.74 / 0.63 vs the engine's 0.20), guess falls and slip rises with difficulty (hard-item slip ≈ 0.64 in both), 4.9% / 22.2% of per-skill fits are degenerate. Every simulator parameter now has a provenance entry (`EVALUATION_PROTOCOL.md` §8). Both datasets are maths, not Python: they set ranges, not values. CSEDM 2019 (Python) is deferred to W8 (download failed). An exploratory S4 world driven by the ASSISTments fit gives adapt = static on goal mastery in calm (86.0% each) with 0 vs 1335 violations.

8. Demo

`frontend/`: FastAPI adapter (9 endpoints, API plans equal direct engine calls) and a React app with setup, diagnostic, curriculum map, plan timeline, "Why?" trace panel, study mode, disruptions, autopilot and evaluation pages; deterministic demo mode; Playwright run of the 6-minute demo script passes. Run: `make demo-test` (see `frontend/README.md`). Demo content is AI-drafted and awaits human review.

9. Limitations and W8 plan

Synthetic evaluation; short horizons; post-hoc scheduler fix (all runs reported); diagnostic not better than the prior in IRT worlds; review effect mixed; reviewed-vs-not retention confounded by selection. W8: ranking policy (ZPDES bandit with logged propensities), Elo/HLR models, CP-SAT scheduler, CSEDM Python fit, longer-horizon evaluation, classroom pilot with the demo.

10. Traceability and reproducibility

All 23 TC IDs with test evidence: `docs/TRACEABILITY.md` (W7 changes: TC2.1 clause restored and reported as partly met; TC3.1 now verified for executed episodes; TC3.2 mid-day; TC4.1 library parity; TC7.1 exactly-one audit). Commands, seeds and code hashes: notes §14; each results file records its `code_hash` (SHA-256 over `src/**/*.py`) and timestamp; pre-registrations in `docs/preregistration/`.